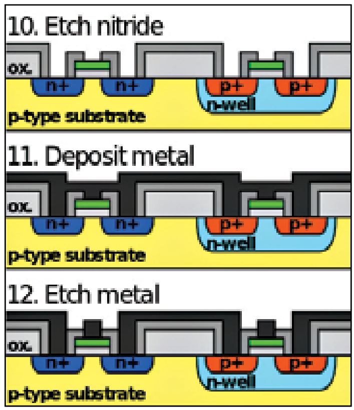
Front-end-of-line(FEOL) processing

This is the process by which the formation of transistors happens on the silicon. The raw wafer, in this case is engineered by growing a highly pure silicon layer that's practically defect-free. This is done by a process called epitaxy. In more advanced logic devices, even before the silicon epitaxy, certain tasks are performed which improve the performance of the transistors that are to be built.

One such method involves introducing a straining step in which a silicon variant like silicon-germanium(SiGe) is deposited. Once that's done, the crystal lattice would become stretched to a certain extent. This would result in a better electronic mobility.

Another method is termed silicon on insulator technology. In this, an insulating layer is inserted between the raw silicon wafer and the thin layer of silicon epitaxy. The advantage of this method is that the transistors created would have lesser parasitic effects.



Deep stuff is FEOL!

Gate oxide and implants

The step next to front-end surface engineering involves growth of the gate dielectric, patterning the gate and also patterning of the source and drain regions. It also involves implantation or diffusion of dopants. This is so that the required complementary electrical properties could be obtained.

In the case of dynamic random-access memory devices, the storage capacitors too are fabricated at this point. They are usually stacked above the access transistor.

Back-end-of-line (BEOL processing)

Metal layers

After the creation of the different semiconductor devices, they need to be interconnected so that the required electrical circuits could be formed. This happens over a series of wafer processing steps that are collectively called BEOL.